A Reproducible Simulation Framework for Comparing Legislative Mechanism Bundles

Stress Tests for Productivity, Revision Moderation, Public Support, and Risk

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Legislatures need enough productivity to act, but productivity alone can also mean volatility, agenda capture, or weakly supported lawmaking. This paper presents a reproducible computational framework for comparing selected legislative mechanism bundles under shared synthetic assumptions. The simulator routes the same generated legislatures and bill streams through legislative-style processes that vary agenda access, proposal revision, voting thresholds, alternative selection, public review, lobbying safeguards, affected-group protection, package bargaining, and reversibility.

The contribution is methodological: a reusable experimental architecture for diagnosing how legislative collective-decision mechanisms shift the relationship among throughput, moderating revision, support, risk, and procedural cost under controlled synthetic worlds. The framework is designed to make tradeoffs inspectable rather than to rank real institutions. It reports a diagnostic dashboard separating productivity, revision moderation, public support, low-public-support enactment, risk control, administrative cost, and generated public benefit per submitted proposal. A main comparison campaign demonstrates how these metrics separate across representative mechanism families, while scripted diagnostics generate seed, ablation, manipulation-stress, chamber-structure, and empirical flow sanity checks. The reported results are synthetic, generator-dependent design hypotheses.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies → Modeling and simulation**; • **Applied computing →** *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, revision moderation, collective decision-making, voting systems

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1 Introduction

The motivating question is whether systematic comparison can distinguish legislative structures that merely pass bills from those that combine productivity, revision, and representative responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because disagreement is suppressed. In this framework, representative responsiveness is operationalized through generated public support, generated public benefit, revision behavior, and burden diagnostics for affected groups.

The central modeling choice is to treat final passage thresholds as only one part of a larger institutional process that includes proposal access, agenda control, revision, review, and correction. Burden-shifting passage rules are included only as stress tests because they expose how proposal access, objection rights, information, and correction interact with final voting thresholds.

This paper treats legislative institutions as collective-intelligence architectures: procedural rules determine who can contribute proposals, which signals are amplified, how expertise and public support enter decisions, how disagreement

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is resolved, and how errors are corrected. The simulator is therefore not a forecast of Congress but a tool for comparing institutional aggregation mechanisms under shared synthetic conditions.

The paper asks three questions. First, can a modular simulator distinguish throughput-producing rules from rules that produce moderated, broadly supported revisions under shared synthetic assumptions? Second, which mechanism families remain relevant when productivity, revision moderation, public support, risk, and administrative burden are evaluated separately? Third, which findings survive sensitivity checks on assumptions, mechanism ablations, and manipulation stressors?

This paper should be read as a framework and stress-test study, not as a validated model of Congress or a recommendation for constitutional reform. The reported results compare institutional mechanisms under shared synthetic assumptions. Empirical flow checks screen for gross implausibility, while validation against district opinion, implementation, court review, campaign finance, and cross-national institutional data remains future work.

2 Related Work

The model draws on individual-level simulation, spatial voting, and institutional theory. Agent-based and computational political models are useful when heterogeneous actors interact under rules that create aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Spatial voting motivates one-dimensional ideal points and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989; Romer and Rosenthal 1978]. Pivotal-politics and veto-player theories motivate comparing simple majority, supermajority, bicameral, and executive-veto systems [Krehbiel 1998; Tsebelis 2002]. Committee and agenda-control theory motivates treating floor denial, referral, information, and amendment control as outcomes rather than missing observations [Cox and McCubbins 2005; Gilligan and Krehbiel 1987; Krehbiel 1991; Shepsle 1979; Shepsle and Weingast 1987; Weingast and Marshall 1988].

Several tested mechanisms come from adjacent literatures. Lobbying is modeled as access, effort, information, public pressure, and participation bias rather than only direct vote buying [Austen-Smith 1993, 1995; de Figueiredo and Richter 2014; Gilens and Page 2014; Hall and Deardorff 2006; Hall and Wayman 1990]. Citizen-panel variants draw on deliberative polling and mini-public practice [Fishkin 2009; Organisation for Economic Co-operation and Development 2020]. Agenda-credit variants echo quadratic-voting and storable-vote work on expressing intensity under scarcity [Casella 2005; Lalley and Weyl 2018]. Reversibility variants draw on temporary legislation and sunset review [Gersen 2007; Ranchordas 2014]. Policy tournaments draw on Condorcet and agenda-manipulation problems [McKelvey 1976; Tideman 1987]. Model documentation follows ODD and ODD+D conventions [Grimm et al. 2010; Müller et al. 2013]. Existing work typically studies one institutional mechanism, one voting rule, or one deliberative process at a time. This paper contributes an executable comparison environment that treats institutional design as a modular collective-intelligence architecture: the same synthetic actors and proposal streams are passed through different agenda, revision, voting, review, and correction mechanisms so diagnostic tradeoffs become inspectable.

3 Simulator

Each run creates a synthetic legislature, bill stream, scalar status quo, and institutional process. Legislators have ideal points $z_i \in [-1, 1]$, party labels, district preferences, and heterogeneous sensitivity to party pressure, constituency pressure, lobbying, reputation, and moderating revision. Bills have a proposer, policy position x_b, public support p_b, generated public benefit g_b, lobby pressure ℓ_b, salience, private gain, issue domain, public-benefit uncertainty, affected-group support, and concentrated harm. Enacted bills update one abstract status quo q_t, so votes evaluate policy movement rather than isolated proposals. Issue domains affect lobbying, harm, committees, and diagnostics, but the baseline model intentionally collapses policy movement into one dimension.

The voting kernel is status-quo-relative. A legislator's spatial utility is $u_i(x) = -(x - z_i)^2$. For bill b, the vote score is a weighted sum of ideology, party, public support, lobbying, revision moderation, and reputation terms:

$$s_i(b) = \sum_k w_k f_k(i, b, q_t) + \epsilon_i, \qquad \Pr(Y_i = 1) = \sigma(\max(-12, \min(12, s_i(b)))).$$

The ideology term uses $u_i(x_b) - u_i(q_t)$, party pressure uses the same comparison at the party position, constituency pressure is proportional to $2(p_b - 0.5)$, lobbying is proportional to ℓ_b, and the revision-moderation term rewards moderation, public legitimacy, and incremental policy movement. The outcome is stochastic but seed-reproducible.

Institutional modules then transform the bill lifecycle:

$$(W_t, b, q_t) \to \text{access} \to \text{revision or selection} \to \text{vote} \to \text{review} \to (q_{t+1}, m_t).$$

Modules can deny access, route to committees, force discharge or open-calendar consideration, spend attention, apply lobbying safeguards, mediate text, select among alternatives, impose harm-weighted thresholds, run citizen or objection review, apply vetoes or judicial review, and register enacted laws for later correction. Post-enactment review tracks implementation delay, capacity, underfunding, nonenforcement, and renewal pressure as separate diagnostics.

The stylized U.S.-like conventional benchmark uses House majority passage, a Senate 60 percent threshold, presidential veto, two-thirds override, and an agenda-access gate. The gate prevents the benchmark from treating every introduced bill as a floor bill; it schedules bills using observable proxies: public support, salience, policy stability, sponsor-party fit, partisan queue conflict, capture risk, concentrated harm, and uncertainty. It does not observe generated public benefit directly.

The baseline generator is moderation-friendly by construction: generated public benefit increases as $|x_b|$ decreases, before noise, private-gain penalties, and adversarial-case modifications. That assumption makes content-selection and moderating-revision mechanisms easier to reward. The results section therefore treats the ordering as a demonstration of framework behavior under the implemented generator, and the appendix reports alternative worlds where extreme proposals can be beneficial, moderation can dilute value, lobbying can supply information, public support can be biased, and majority support can coincide with concentrated harm.

4 Metrics and Campaign

The simulator reports a dashboard rather than a single welfare score. Table 2 defines the metrics used in the main paper; secondary diagnostics remain in generated reports and appendix tables.

Yield prevents a system that blocks nearly all proposals from looking equivalent to one that produces generated public value. Risk control aggregates low-support passage, minority harm, lobbying capture, public-preference distortion, concentrated-harm passage, proposer gain, and policy shock; administrative feasibility is the inverse of the procedural-load index.

As one sensitivity check, the paper reports a revision-moderation-weighted value profile combining revision moderation, productivity, risk control, administrative feasibility, and public support. The main interpretation relies on metric-level tradeoffs rather than this scalar score.

All main results come from one reported comparison campaign. It combines broad assumptions, adversarial generator cases, weighted party-system cases, and a stylized rising-contention timeline in one CSV. The canonical run uses 120

Table 1. Key generator and vote-kernel assumptions visible in the main paper. Exact formulas, bounds, scenario constants, and weights are reported in the appendix and code. These are modeling choices, not empirical estimates.

Component	Core modeling assumption	Why it matters
Legislator ideal points	One-dimensional, faction-centered distribution with heterogeneous party, constituency, lobbying, reputation, and revision-moderation sensitivities.	Determines ideological conflict and makes moderation mostly movement along one policy line.
Status quo and bills	Bills are proposer-centered policy movements evaluated against a scalar status quo; enacted bills update that status quo.	Makes voting status-quo-relative, but collapses multidimensional politics into one abstract dimension.
Public benefit	Baseline g_b is moderation-linked: $E[g_b] \approx 0.34{+}0.34(1{-}\lvert x_b \rvert){-}0.08\,private_b$, before noise and adversarial-world changes.	This is the most important generator vulnerability because it can advantage content-selection and moderating-revision mechanisms.
Public support	Support is synthetic and correlated with generated benefit, lobbying pressure, private gain, and noise.	Public-support diagnostics are not observed public opinion and inherit generator assumptions.
Lobby/private gain	Lobby pressure is partly tied to private gain, with alternative worlds where lobbying can provide useful information.	Drives capture diagnostics and makes anti-lobbying assumptions visible.
Harm/uncertainty	Concentrated harm and uncertainty rise with distance, salience, private gain, signal mismatch, lobbying, and noise.	Drives risk screens, harm-weighted thresholds, and low-support/risk-control metrics.
Vote kernel	Legislators combine status-quo-relative spatial utility, party pressure, public support, lobbying, a revision-moderation term, reputation, and seeded noise.	The revision-moderation score is a model proxy, not a neutral moral measure of good bargaining.

Table 2. Main metrics and interpretation.

Metric	Direction	Definition	Caveat
Productivity	↑	Share of submitted bills enacted.	Can reward bad throughput.
Yield	↑	Generated public benefit per submitted proposal.	Synthetic and generator-dependent.
Revision moderation	↑	Score combining moderation, legitimacy, and incrementalism.	Not a neutral welfare measure.
Public support	↑	Enacted-bill support composite from generated support signals.	Not observed public opinion.
Low-support enactment	↓	Enacted bills below 50% generated support.	Depends on synthetic support threshold.
Risk control	↑	Inverse composite of harm, capture, low support, proposer gain, and policy shock.	Composite score.
Admin cost	↓	Procedural-load index from attention, review, challenge, amendment, and selection costs.	Indirect cost proxy.

generated worlds per case, 101 legislators, 60 bills per run, and seed 20260428. Cases vary polarization, party loyalty, revision-moderation culture, lobbying, constituency pressure, party count, proposal pressure, capture pressure, and anti-lobbying reform demand. The adversarial bill-stream cases stress extremity, capture, delayed benefits, concentrated harm, and backlash dynamics.

Scenario selection follows a fixed coverage rule: include the stylized U.S.-like benchmark, simple majority, committee regular order, discharge petition, open floor calendar, a combined content-selection family, anti-capture access, harm-weighted majority, one burden-shifting stress case, one burden-shifting mediation case, and a portfolio hybrid. The appendix reports the broader scenario catalog and fixed-rule family champions.

The chamber-structure supplement tests representation architecture, committee power, eligibility filters, and ex ante review. It is separate because it asks how those structures shift the same productivity/revision-moderation tradeoff.

Empirical flow sanity checks are artifact screens, not calibration in the statistical sense. `make calibrate` compares conventional baselines with broad benchmark ranges derived from Voteview, Congress.gov, govinfo, Comparative Agendas Project, ParlGov, lobbying-disclosure and OpenFEC records, Center for Effective Lawmaking measures, and

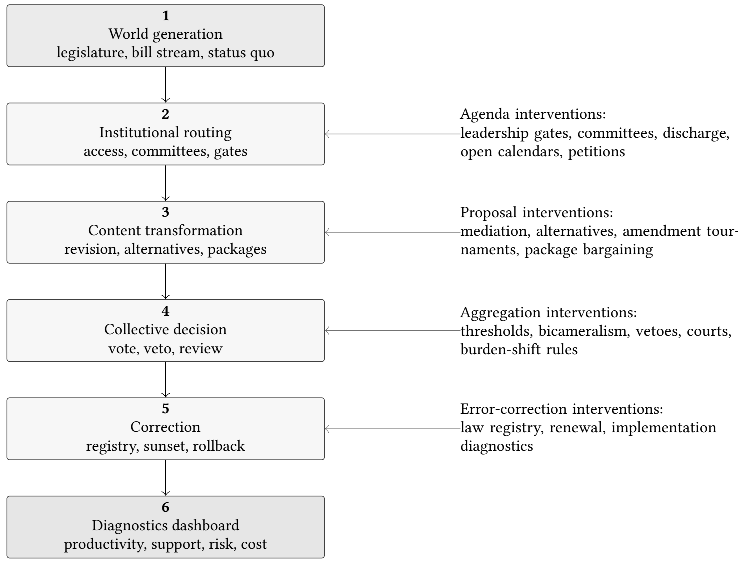


Fig. 1. Modular simulation pipeline. Each institutional bundle intervenes at one or more stages while generated legislatures, bill streams, and status-quo-relative voting remain shared across comparisons.

Table 3. Mechanism families used in the main comparison. Labels match the Results figures; the appendix reports the broader implemented catalog.

Family	Labels	Mechanism theme	Primary stress risk
Conventional benchmark	CUR	House/Senate/veto gate	Bottlenecks
Simple majority	SM	Affirmative threshold	Low screening
Committee regular order	COMM	Committee-first gate	Burial/capture
Discharge petition	DIS	Committee bypass	Bypass pressure
Open floor calendar	OPEN	Default-open agenda	Floor overload
Content selection	SEL	Alternatives/tournament	Clones/agenda manipulation
Anti-capture access	ACG	Pre-floor capture screen	Gate bias
Harm-weighted majority	HARM	Affected-group threshold	Holdouts
Burden-shifting stress	DP	Pass unless blocked	False silence
Burden shift + mediation	DPM	Mediation plus challenge	Cost/weak consent
Portfolio hybrid	PORT	Combined safeguards	Complexity

V-Dem indicators [Comparative Agendas Project 2026; Döring and Manow 2024; Federal Election Commission 2026; Lewis et al. 2021; Library of Congress 2026; United States Government Publishing Office 2020; United States Senate Office of Public Records 2026; V-Dem Institute 2026; Volden and Wiseman 2014]. Separate flow checks use six documented empirical comparison samples: a 180-row Congress.gov bill-progression extract, a Voteview roll-call extract, a Senate LDA lobbying-disclosure extract, and Congress.gov-derived topic, sponsor, and committee summaries. These support bill attrition, floor load, committee reporting, coalition size, party unity, topic throughput, sponsor concentration, and lobbying-spend concentration checks.

Table 4. Empirical flow sanity checks for the conventional benchmark. Raw inputs are optional and do not validate the synthetic welfare, revision moderation, harm, capture, or public-support metrics.

Signal	Raw	Simulator proxy	Does not validate
Bill attrition	enactmentRate=0.017	current-system/productivity	public benefit or law quality
Floor consideration	floorLoad=0.122	current-system/floor	agenda fairness or policy merit
Committee reporting from bill actions	committeeReportRate=0.083	current-system/floor	committee expertise or capture
Roll-call coalition size	coalitionSize=0.604	current-system/averageEnactedSupport	public opinion
Party unity	partyUnity=0.946	current-system/averageEnactedSupport	public support or legitimacy
Sponsor proposal concentration	sponsorIntroductionGini=0.389	current-system/proposerAccessGini	capture or representational equity

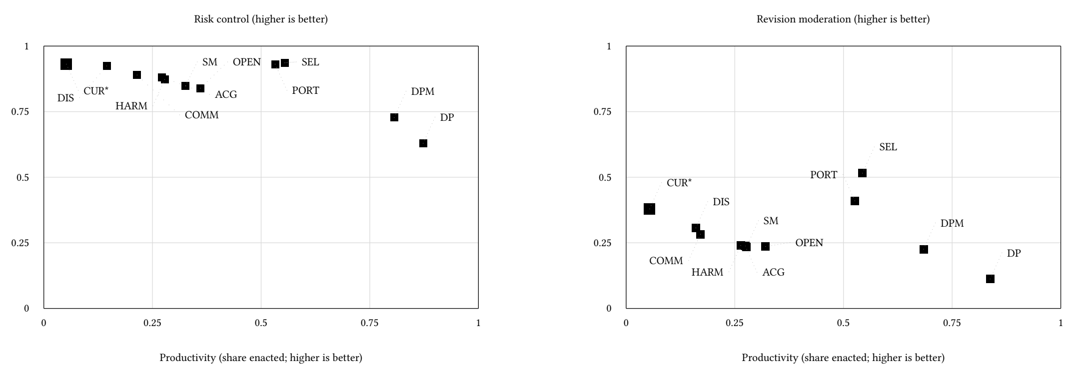
5 Results

The selected mechanism-family comparison shows three patterns. First, throughput and support/risk diagnostics separate: mechanisms that enact more proposals can also enact more low-public-support proposals. Second, content-selection stages change the collective-intelligence process by altering which alternatives enter final aggregation, not merely by changing the final yes/no threshold. Because pairwise alternatives and amendment tournaments produce nearly identical aggregate profiles in this campaign, the main text reports them as a single content-selection family. Third, portfolio safeguards reduce several risk diagnostics but increase procedural load. The stylized U.S.-like benchmark has low productivity but stronger revision moderation, risk, and support diagnostics than its directional profile suggests.

The empirical flow sanity checks are boundary checks for observable legislative-flow proxies. They do not validate generated public benefit, revision moderation, harm, capture, or public-support diagnostics.

Detailed numeric results, Pareto checks, ablation deltas, manipulation-stress losses, and separate PAIR/AMT rows appear in the appendix. The generator-sensitivity rows and fairness controls show that the framework can expose dependence on generator assumptions, but they do not remove that dependence. These results are mechanism-behavior hypotheses under the implemented generator, not general rankings of legislative institutions.

The chamber-structure supplement is reported in the appendix. It asks a separate question about representation architecture and review design, so it is not part of the ACM paper's central evidence chain.



(a) Risk control (b) Revision moderation

Fig. 2. Productivity tradeoff views for the representative systems. Panel (a) plots risk control and panel (b) plots revision moderation under weighted party-system sensitivity cases; higher is better on both y-axes. Labels are the mechanism-family labels from Table 3. The asterisk and larger square mark the stylized U.S.-like benchmark.

The burden-shifting stress cases illustrate the same separation. Open burden-shifting passage has high productivity but weak revision-moderation and support diagnostics, while mediation improves some content metrics at additional procedural cost. These results are best read as failure-mode probes rather than as a central reform comparison.

Manipulation-stress cases use bounded adversaries rather than fully co-evolving strategic parties. The adversary can choose clone or decoy proposals, noisier citizen-panel information, loose harm claims, astroturf objection pressure, proposal flooding, or anti-lobbying backlash under fixed information and budget assumptions. Stress objectives are enactment of low-support bills, blockage of high-benefit bills, or administrative overload. Appendix tables report paired losses, while adaptive worst-case search remains future work.

The portfolio variants illustrate the tradeoff between combining safeguards and increasing administrative load. They improve several risk diagnostics but do not dominate simpler alternative-selection mechanisms.

Table 5. Selected metric profiles for the main mechanisms.

Label	Prod.	Rev. mod.	Public	Risk	Admin	Example profile
CUR*	**0.05**	**0.36**	**0.77**	**0.93**	**0.98**	**0.57**
SM	0.33	0.24	0.68	0.85	0.93	0.55
COMM	0.22	0.28	0.73	0.89	0.97	0.56
DIS	0.14	0.31	0.75	0.92	0.92	0.56
OPEN	0.36	0.25	0.67	0.84	0.90	0.55
SEL	0.55	0.50	0.79	0.94	0.71	0.67
ACG	0.27	0.25	0.70	0.88	0.94	0.55
HARM	0.28	0.26	0.69	0.87	0.93	0.55
DP	0.87	0.14	0.57	0.63	0.93	0.57
DPM	0.81	0.24	0.62	0.73	0.79	0.59
PORT	0.53	0.43	0.76	0.93	0.76	0.65

Note: Values are normalized to the metric direction shown in Table 2; grayscale shading is secondary. Admin is administrative feasibility, the inverse of the cost index. The asterisk marks the stylized U.S.-like benchmark.

The reported ordering should therefore be read as a hypothesis about mechanism behavior under the implemented generator, not as an institutional ranking.

6 Limitations

The model is stylized. It uses one primary policy dimension, generated legislators, generated support and benefit, simplified issue domains, and abstract lobbying units. The multidimensional package scenario is a cross-issue bargaining proxy, not a full two-dimensional ideal-point model; claims about alternatives, tournaments, revision moderation, and package bargaining are therefore intentionally narrow. Public-support diagnostics distinguish salience and costly affected-group pressure from cheap amplified volume, but they are still proxies. Because public benefit is internal to the generator, welfare comparisons are conditional on model assumptions. Systems that screen extreme, uncertain, or lobby-heavy bills may look better partly because the generator often encodes those features as lower expected public value. Adversarial cases partially relax this assumption, but welfare remains a synthetic signal.

Table 6. Robustness and failure-mode summary. Raw numeric rows are in the appendix.

Probe family	Main pattern changes?	Main implication
Extreme-beneficial reform	Yes	Portfolio has the highest displayed profile and content selection remains close; this directly tests the moderation-friendly benefit generator.
Revision dilution	Yes	Portfolio has the highest displayed profile when moving toward the center can reduce generated value.
Lobbying information	No major change in displayed pattern	The content-selection pattern persists when organized lobbying can carry useful information.
Public-opinion error	Partial	Portfolio and content selection are near-tied when support and benefit decouple.
Minority-rights harm	No major change in displayed pattern	The content-selection pattern persists, but the case targets majority-supported concentrated harm.
Cost-constrained fairness	Unresolved	Reported fairness rows add conventional revision, mediation, and committee amendment; they do not yet hard-budget process cost.
Ablation/stress probes	Failure modes	No alternatives weakens the content-selection result; clone/decoy, astroturf, and loose-harm cases expose manipulability.
Objective-set frontier	Yes	Adding administrative feasibility or harm/support objectives changes which systems are non-dominated.

The empirical layer screens conventional baselines against empirical ranges, and the flow sanity checks cover six empirical inputs. That is enough to make several agenda-access and legislative-flow proxies inspectable, but it is not enough for model validation or parameter fitting. District opinion, campaign finance, court review, implementation feedback, law revision, and cross-national institutional comparison remain validation gaps; findings that depend on those components should be read as synthetic design hypotheses. The committee and sponsor summaries are derived from sampled Congress.gov bill actions and are especially rough; fixture samples exercise adapter shapes but are not validation data. Strategic behavior remains bounded: proposers and lobby groups adapt, and norm erosion can raise delay, but parties, committees, coalitions, courts, agencies, media, and elections do not yet co-evolve. Chamber scenarios are sensitivity prototypes. Citizen panels, initiatives, objection windows, tournaments, audits, challenge tokens, judicial review, and law-registry review are optimistic prototypes whose costs are represented only indirectly. Package bargaining models side payments, delay, jurisdictional trades, and distributive load, not legal text or appropriations.

7 Reproducibility

The submitted manuscript is anonymized for double-blind review. The artifact is a Java 21 project with Makefile entrypoints and vendored/generated CSV summaries for the paper workflow. `make reproduce-paper-offline` is the no-network reproduction target; it regenerates the campaign, diagnostics, figures, PDFs, word-count check, anonymity scan, figure-label and table consistency checks, and PDF manifest from fixed seeds. `make supplement-anonymous` builds the double-blind supplement. Optional fixture samples can be refreshed with `make fetch-validation-samples`; they remain separate from raw empirical inputs. `make validation-gap-report` writes the empirical-boundary report and appendix table. `make build-bill-progression-raw` rebuilds the Congress.gov bill-progression sample, and `make build-core-raw-validation` rebuilds the broader Voteview, Congress.gov-derived, and LDA empirical comparison samples when network access and any required API key are available.

The minimal local smoke test is `make test`, which compiles the simulator and runs the Java test suite in under one minute on the authoring workstation. The full paper workflow typically takes several minutes because it reruns campaign, diagnostic, figure, and PDF generation. Reproduced values should match the tracked CSV/PDF manifest under the fixed seeds; raw external-data rebuilds are optional and intentionally outside the no-network path.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. OpenAI ChatGPT/Codex tools available in the authoring environment during the April–May 2026 revision period were used for grammar and style editing, restructuring suggestions, code debugging assistance, and drafting assistance for portions of the simulator implementation, generated-report scripts, and manuscript prose. Tool outputs were edited and checked by the author; references, code outputs, experiments, tables, and claims were manually reviewed against the repository artifacts before inclusion. No large language model is listed as an author.

8 Conclusion

This paper contributes a reproducible simulation framework for comparing legislative mechanism bundles as collective-intelligence architectures. The reusable contribution is an experimental architecture for making legislative collective-decision tradeoffs inspectable under shared synthetic worlds. The results support the framework's main methodological claim: legislative productivity, revision moderation, public support, risk control, and administrative burden separate under different institutional mechanisms. In the reported synthetic campaigns, content-selection stages often change the productivity/risk surface more than threshold-only baselines, while portfolio variants illustrate the cost of combining safeguards. Future work should replace synthetic proxies with stronger public-opinion, lobbying, court-review, implementation, and bargaining data.

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